

Shot Answer questions

1) The cost price of 30 Apples equals the selling price of 25 Apples.

Find the profit or loss percent.

$$SP \frac{30}{6} = \frac{25}{5} CP$$

$$SP 6 = 5CP$$

$$SP = 5 \quad CP = 5$$

$$L = CP - SP \\ = 6 - 5 = 1$$

$$Loss\% = \frac{L}{CP} \times 100$$

$$= \frac{1}{6} \times 100$$

$$= 16\frac{2}{3}\%$$

2) Find the sum of factors of 200?

$$\begin{array}{r} 2 \overline{) 200} \\ 2 \overline{) 100} \\ 2 \overline{) 50} \\ 2 \overline{) 25} \\ 5 \overline{) 25} \\ 5 \end{array}$$

$$2 \times 2 \times 2 \times 5 \times 5 \Rightarrow 2^3 \times 5^2 \\ \Rightarrow (2^0 + 2^1 + 2^2 + 2^3) (5^0 + 5^1 + 5^2) \\ = 15 \times 31 = 465$$

3) Find the LCM and HCF of 42, 63 & 140?

$$\begin{array}{r} \text{LCM } 7 \overline{) 42, 63, 140} \\ 2 \overline{) 6, 9, 20} \\ 3 \overline{) 3, 9, 10} \\ 1 \quad 3 \quad 10 \end{array}$$

$$7 \times 2 \times 3 \times 3 \times 10 \\ = 1260$$

$$\text{LCM } 1260$$

$$7 \overline{) 42, 63, 140} \\ 6, 9, 20$$

$$\text{HCF} = 7$$

4) The ratio of two numbers is 15:11. If the HCF of those two numbers is 13, find the LCM of those two numbers.

$$\text{HCF } 15x, 11x$$

$$x \overline{) 15x, 11x} \\ 5, 11$$

$$HCF = x = 13$$

$$\begin{array}{r} 5x \quad 11x \\ 5 \times 13 \quad 11 \times 13 \\ \hline 195 \quad 143 \end{array}$$

$$\text{LCM } 13 \overline{) 195, 143} \\ 15, 11$$

$$165 \times 13$$

$$= \underline{\underline{2145}}$$

5) At what rate percent per annum will Rs. 2000 earn Rs.1000 simple interest in 5 years?

$$R = \frac{SI \times 100}{P \times T} = \frac{1000 \times 100}{2000 \times 5} = 10$$

$$\boxed{R = 10\%}$$

6) What is the remainder when 7^{84} is divided by 342?

$$\frac{(7^{84})}{342} = (7^3)^{28} = (343)^{28}$$

$$\frac{(343)^{28}}{342} = (1)^{28} = 1$$

7) What is 40% of 30% of 800?

$$\frac{40}{100} \times \frac{30}{100} \times 800$$

$$4 \times 3 \times 8 = 96$$

8) What is 30% of 25% of 480?

$$\frac{30}{100} \times \frac{25}{100} \times 480 = 36$$

9) If CP of 15 articles is equal to SP of 12 articles. Find the profit or loss percent

$$CP\ 15 = SP\ 12$$

$$CP = 12$$

$$SP = 15$$

$$P = SP - CP = 15 - 12 = 3$$

$$P\% = \frac{P}{CP} \times 100 = \frac{3}{12} \times 100 = \frac{1}{4} \times 100 = 25\%$$

10) If the simple interest on Rs. 9000 for 3 years is Rs. 2700, what is the rate of interest per annum?

$$SI = \frac{PTR}{100} \Rightarrow 2700 = \frac{9000 \times 3 \times R}{100}$$

$$R = 10\%$$

$$\boxed{R = 10\%}$$

11) What is the remainder when 2^{60} is divided by 5?

$$\begin{aligned} \frac{2^{60}}{5} &= (2^2)^{20} = (3 \times 2^4)^{20} \\ &= \frac{4^4}{5} \\ &= \frac{3 \times 32}{5} = \frac{96}{5} \\ &= 1 \end{aligned}$$

12) What is the remainder when 17^{200} is divided by 18?

$$\frac{(17)^{200}}{18} = (-1)^{200} = 1$$

13) 40% of 265 + 35% of 180 = ?

$$\left(\frac{40}{100} \times 265\right) + \left(\frac{35}{100} \times 180\right)$$

$$106 + 63 = 169$$

14) Find the cost price of an object, when the selling price is Rs. 1950 and the loss is 25%.

$$100 - 25 = 75\%$$

$$sp = 75\% \rightarrow 1950$$

$$cp = 100\% \rightarrow ?$$

$$\frac{100}{75} \times 1950$$

$$= 650 \times 4 = \underline{\underline{2600}}$$

15) A sum of Rs. 8000 amount to Rs. 9200 in 3 years at simple interest what is the rate of interest percent per annum?

$$SI = \frac{P \times T \times R}{100}$$

$$1200 = \frac{8000 \times 3 \times R}{100}$$

$$R = 5\%$$

$$\boxed{R = 5\%}$$

Long Answer questions

1) A painter needs 30 kg of paint to cover a building. Allowing for 15% wastage, and if the paint is sold in 3 kg cans at Rs. 25 per can, what will be the total cost of the paint purchased?

$$\begin{aligned} &= 30 \text{ kg} + 15\% \text{ of } 30 \\ &= 30 \text{ kg} + \frac{15}{100} \times 30 \\ &= 30 + 4.5 \\ &= 34.5 \end{aligned}$$

2) At Green Valley School, there are 750 students in total. If 40% of the students are girls, find the number of boys and girls in the school.

$$\begin{array}{l|l} 100\% \rightarrow 750 & 100 \rightarrow 750 \\ \text{Girls} = 40\% & 60\% \rightarrow ? \\ \text{boys} \Rightarrow 60\% & \\ \hline 100\% \rightarrow 750 & \frac{60}{100} \times 750 = 450 \\ 40\% \rightarrow ? & \\ \frac{40}{100} \times 750 = 300 & \end{array}$$

3) A shop earned a total profit of Rs. 5400 in a month. The owner decides to share the profit among three employees in the ratio 3: 4: 5 according to their performance. How much will the second employee receive?

$$\begin{aligned} \text{profit} &= 5400 \\ \text{Total parts} &= 3x + 4x + 5x = 12x \\ \text{one part} &= x = \frac{5400}{12} = 450 \\ \text{The second employee} &= 4x \\ 4 \times 450 &= 1800 \end{aligned}$$

4) Three traffic lights at a junction change at regular intervals. The first light changes every 30 seconds, The second light every 40 seconds, and the third light every 50 seconds. If all the lights change together at 8:00 AM, after how many seconds will they change together again?

$$\text{LCM } 10 \overline{) 30, 40, 50}$$

$$\begin{aligned} \text{LCM} &= 10 \times 3 \times 4 \times 5 \\ &= 600 \text{ seconds} \\ &= 10 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{After} &= 8 \text{ AM} + 10 \text{ m} \\ &= 8:10 \text{ AM} \end{aligned}$$

5) There are three equal glasses filled with milk and water. Each glass contains the following proportion of milk to water, first glass 2:3, in the second as 3:5 and in the third 4:1, The contents of the three glasses are emptied into a single vessel. What is the proportion of milk and water in the new vessel?

$$\begin{array}{ccc} \boxed{2:3} & \boxed{3:5} & \boxed{4:1} \\ \frac{2}{5} \quad \frac{3}{5} & \frac{3}{8} \quad \frac{5}{8} & \frac{4}{5} \quad \frac{1}{5} \\ \frac{2}{5} + \frac{3}{8} + \frac{4}{5} & \frac{3}{5} + \frac{5}{8} + \frac{1}{5} & \end{array}$$

$$\begin{array}{l} 16 + 15 + 32 \quad 24 + 25 + 18 \\ 63 : 57 \\ 4 : 19 \end{array}$$

6) What is the Rational form of 45. 124512451245...?

$$\begin{aligned} 45. \overline{1245} \\ 45 + 0.\overline{1245} \\ 45 + \frac{1245}{9999} \end{aligned}$$

$$\frac{449955 + 1245}{9999} = \frac{451200}{9999}$$

7) What is the Rational form of 12.123123123

$$\begin{aligned} &= 12.\overline{123} \\ &= 12 + 0.\overline{123} \\ &= 12 + \frac{123}{999} \end{aligned}$$

$$\begin{aligned} &= \frac{11988 + 123}{999} \\ &= \frac{12111}{999} \end{aligned}$$

8) What is the Rational form of $48.178917891789\dots$?

$$\begin{array}{l} 48.\overline{1789} \\ 48 + 0.\overline{1789} \\ 48 + \frac{1789}{9999} \end{array} \quad \left| \quad \begin{array}{l} \frac{(48 \times 9999) + 1789}{9999} \\ \frac{(479952) + 1789}{9999} = 48\overline{1741} \end{array} \right.$$

9) Three bells ring at a temple at regular intervals. The first bell rings every 6 seconds, The second bell every 9 seconds, The third bell every 12 seconds. If all three ring together at the same time initially, how many times will they ring together in 1 hour?

$$\begin{array}{l} \text{LCM } 3 \mid 6, 9, 12 \\ \quad 2 \mid 2, 3, 4 \\ \quad \quad 1, 3, 2 \\ 2 \times 3 \times 3 \times 2 = 36 \text{ seconds} \end{array}$$

$$\begin{aligned} 1 \text{ Hour} &= \frac{60 \times 60}{36} + 1 \\ &= 100 + 1 \\ &= 101 \end{aligned}$$

10) Find the number of trailing zeroes in $300!$

$$\begin{array}{r} 5 \mid 300 \\ 5 \mid 60 \\ 5 \mid 12 \\ 5 \mid 2 \\ \hline 74 \end{array}$$

11) Find the number of trailing zeroes in $180!$

$$\begin{array}{r} 5 \mid 180 \\ 5 \mid 36 \\ 5 \mid 7 \\ 5 \mid 1 \\ \hline 36 \\ 8 \\ \hline 44 \end{array}$$

12) Find the number of trailing zeroes in 850!

$$\begin{array}{r} 5 \overline{) 850} \times \\ \underline{5} \\ 350 \\ 5 \overline{) 350} \\ \underline{35} \\ 0 \\ 5 \overline{) 34} \\ \underline{30} \\ 4 \\ 5 \overline{) 40} \\ \underline{40} \\ 0 \end{array} \quad \underline{\underline{211}}$$

13) A hall requires 50 kg of paint. Accounting for 12% wastage, and if the paint is sold in 5 kg cans at Rs. 40 per can, what will be the total cost of paint purchased?

$$= 50 \text{ kg} + 12\% \text{ of } 50 \text{ kg}$$

$$= 50 + \frac{12}{100} \times 50$$

$$= 56$$

$$\text{Each can} = 5 \text{ kg}$$

$$\text{No. of cans} = \frac{56}{5} = 11.2$$

$$\text{Total Amount} = 11.2 \times 40 = 448$$

14) Three bells commence tolling together and toll at intervals of 6, 8, and 15 seconds respectively. In 60 minutes, how many times do they toll together?

$$\underline{\underline{\text{LCM}}}, \quad 6, 8, 15$$

$$\begin{array}{r} 3 \overline{) 6, 8, 15} \\ \underline{2} \\ 2, 8, 5 \\ 2 \overline{) 2, 8, 5} \\ \underline{2} \\ 0, 6, 5 \\ 2 \overline{) 0, 6, 5} \\ \underline{0} \\ 0, 3, 5 \\ 3 \overline{) 0, 3, 5} \\ \underline{0} \\ 0, 1, 5 \\ 5 \overline{) 0, 1, 5} \\ \underline{0} \\ 0, 0, 5 \\ 5 \overline{) 0, 0, 5} \\ \underline{0} \\ 0, 0, 0 \end{array}$$

$$3 \times 2 \times 4 \times 5 = 120 \text{ seconds} \\ = 2 \text{ minutes}$$

$$\text{In } 60 \text{ min} \Rightarrow \frac{60}{2} + 1 = 30 + 1 = 31$$

15) A company earned a total profit of Rs. 4800 this month. The owner decides to share the profit among three employees in the ratio 5: 3: 2 according to their contribution. How much will the first employee receive?

$$\text{profit} = 4800$$

$$\text{Total parts} = 5x + 3x + 2x = 10 \text{ parts}$$

$$\text{one part} = \frac{4800}{10}$$

$$x = 480 = 480$$

$$\begin{aligned} 5x &= 5 \times 480 \\ &= 2400 \end{aligned}$$